Jiangtao Kong

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Education

William & Mary, Williamsburg, VA — 09/2021 – Present
Ph.D. in Computer Science — Expected Aug 2027
M.S. in Computer Science — 06/2023

Beijing University of Technology, Beijing, China — 09/2015 – 07/2019
B.Eng. in Electronic Information Engineering

Research Interests

- Continual Learning, Multimodal Alignment, Model-Based Reinforcement Learning, Humanoid Control

Selected Publications & Manuscripts

- **FastWorldPlanner: Advantage-Aware Action-Chunk Policy for Online Planning**.
  Jiangtao Kong, et al. (*Submitted to ICLR 2027*)
- **ECA: Efficient Continual Alignment for Open-Ended Image-to-Text Generation**.
  Jiangtao Kong, et al. (*ICML 2026*)
- **WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems**.
  Yuchen Wang*, **Jiangtao Kong***, et al. (*ICML 2026 Spotlight*)
- **ODESteer: A Unified ODE-Based Steering Framework for LLM Alignment**.
  Hongjue Zhao*, Haosen Sun*, **Jiangtao Kong**, et al. (*ICLR 2026*)
- **YoooP: You Only Optimize One Prototype per Class for Non-Exemplar Incremental Learning**.
  Jiangtao Kong, et al. (*TMLR 2025*)
- **Hybrid Memory Replay: Blending Real and Distilled Data for Class Incremental Learning**.
  Jiangtao Kong, et al. (*arXiv 2024*)

*Equal contribution.

Research Experience

World Models and Reinforcement Learning — 2025 – Present

Doctoral Research William & Mary

- **WestWorld.** Co-developed a knowledge-encoded trajectory world model for dynamics prediction and model-based control across diverse robotic systems.
- **SWAN.** Developed action-chunk policies and advantage-guided proposal optimization for efficient online planning, using 50% fewer MPPI refinement iterations across 31 continuous-control tasks.
- **Ongoing: Continual RL.** Investigating humanoid control through sequential task learning, focusing on skill transfer, sample efficiency, and retention of previously learned skills.

JPMorgan Chase & Co., New York, NY — 06/2024 – 08/2024; 09/2024 – 12/2024

Summer Research Associate / Part-time Researcher Leader: Shaohan Hu

- Proposed **ECA** for continual alignment in pretrained vision-language models, combining query mixing, Fisher-guided adapter expansion, and dictionary replay.
- Built four topic-shifted image captioning and VQA benchmarks to evaluate adaptation and forgetting in open-ended image-to-text generation.

Institute of Automation, Chinese Academy of Sciences, Beijing, China — 10/2020 – 06/2021

Research Intern Leader: Jun Wan

- Conducted research on person/vehicle re-identification and face detection, including model adaptation and deployment for computer vision systems.